

Week 9 – Learning Task 2

Problem–Cause–Solution Matching Worksheet

Name: _____

Date: _____

Part A – Matching (Engineering Scenarios)

Instructions:

Match each Problem (A–H) with the correct Cause (1–8) and Solution (a–h).

Write your answers in the table below.

Problems

- A. The hydraulic press failed to build pressure.
- B. The CNC machine stopped during operation.
- C. The power supply unit overheated.
- D. The semiconductor cleanroom particle count increased.
- E. The 3D printer produced uneven layers.
- F. The conveyor belt would not start.
- G. The laboratory oscilloscope displayed no signal.
- H. The cooling fan produced excessive noise.

Causes

- 1. The input fuse was blown.
- ~~2.~~ The hydraulic fluid level was too low.
- ~~3.~~ The HEPA filter was clogged.
- ~~4.~~ The signal probe was not properly connected.
- ~~5.~~ The motor drive belt was misaligned.
- ~~6.~~ The ventilation openings were blocked by dust.
- ~~7.~~ The emergency stop button was activated.
- ~~8.~~ The print bed was not calibrated correctly.

Solutions

- ~~a)~~ Refilled the hydraulic fluid and checked for leaks.
- ~~b)~~ Cleaned the ventilation area and ensured proper airflow.
- ~~c)~~ Realigned the drive belt and tightened the mounting screws.
- ~~d)~~ Reset the emergency stop and restarted the system.
- ~~e)~~ Reconnected the probe and adjusted the signal settings.
- f) Replaced the fuse and verified voltage stability.
- ~~g)~~ Recalibrated the print bed and performed a test print.
- ~~h)~~ Replaced the HEPA filter and retested air quality.

Answer Table

	Problem	Cause	Solution
A	The hydraulic press failed to build pressure.	2	a
B	The CNC machine stopped during operation.	1	f
C	The power supply unit overheated.	6	b
D	The semiconductor cleanroom particle count increased.	3	h
E	The 3D printer produced uneven layers.	8	g
F	The conveyor belt would not start.	5	c

G	The laboratory oscilloscope displayed no signal.	4	e
H	The cooling fan produced excessive noise.	7	d

Part B – Writing Logical Connections

Using signal words (**due to, was caused by, resulted in, therefore, as a result, consequently**), write four complete sentences connecting the problem, cause, and solution.

Example:

The machine failed to initialize **due to** a loose sensor cable. The issue was resolved by reseating the cable and restarting the system.

1.

2.

3.

4.

Language Reminder

Use:

Causatives: was caused by / resulted from / was due to

Conditionals: If the cable is disconnected, the system will not boot.

Action verbs: adjust / replace / inspect / reset / recalibrate / realign